

Spinning or Explaining?

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Outline

The shape of an evolutionary explanation

The fingerprint of evolution

The utility of simulacra

Outline

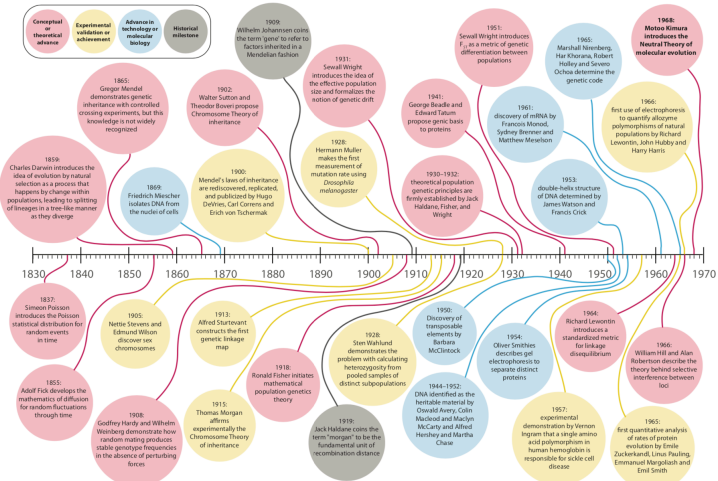
The shape of an evolutionary explanation

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The shape of an evolutionary explanation

The long road to modern evolution



The shape of an evolutionary explanation

The modern concept of evolution

Life on earth evolved gradually beginning with one primitive species — perhaps a self-replicating molecule — that lived more than 3.5 billion years ago; it then branched out over time, throwing off many new and diverse species; and the mechanism for most but not all of evolutionary change is natural selection.

Coyne (2009) *Why Evolution is True*

- ▶ Evolution as such
- ▶ Speciation
- ▶ Natural selection
- ▶ Gradualism
- ▶ Common ancestry
- ▶ Other mechanisms

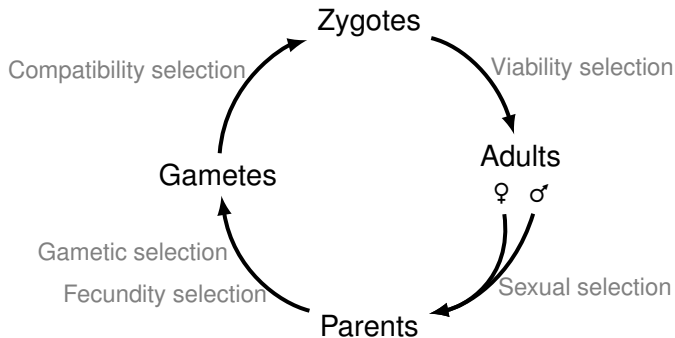
The elements of evolution

Natural selection

- ▶ If a population shows a high potential growth rate
- ▶ And if resources are limited
- ▶ Then only a subset of the population will survive and reproduce (*struggle for existence*)

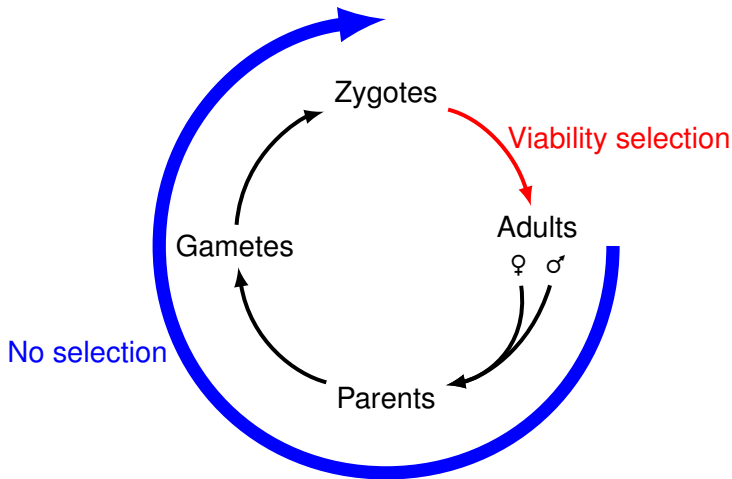
The elements of evolution

Acts at multiple points throughout the life cycle



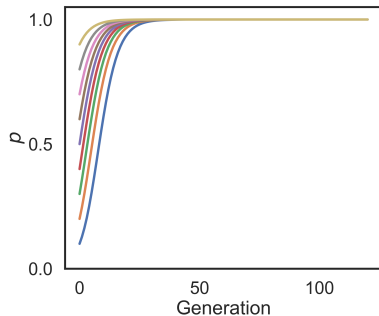
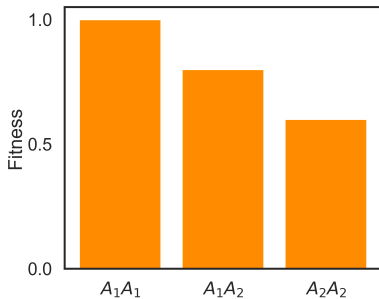
The elements of evolution

Viability selection



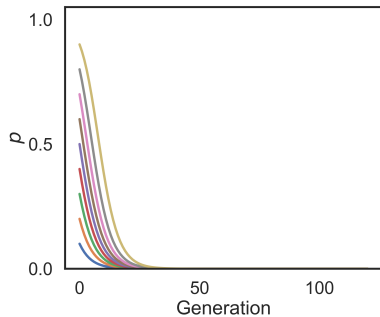
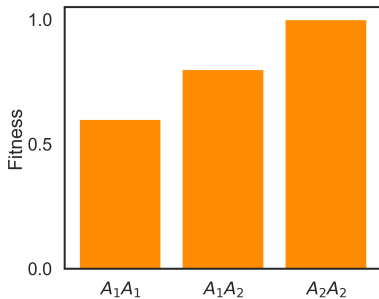
Natural selection

The shape of an evolutionary explanation



Natural selection

The shape of an evolutionary explanation



Genetic drift

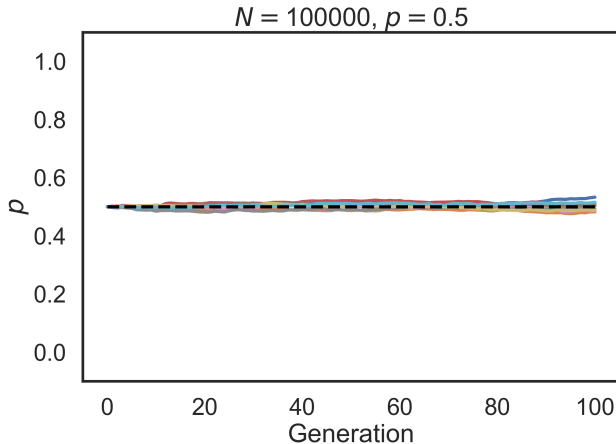
The shape of an evolutionary explanation

How many A_1 alleles will be in the next generation?

$$P(i = A_1) = \binom{2N}{i} p^i q^{2N-i}$$

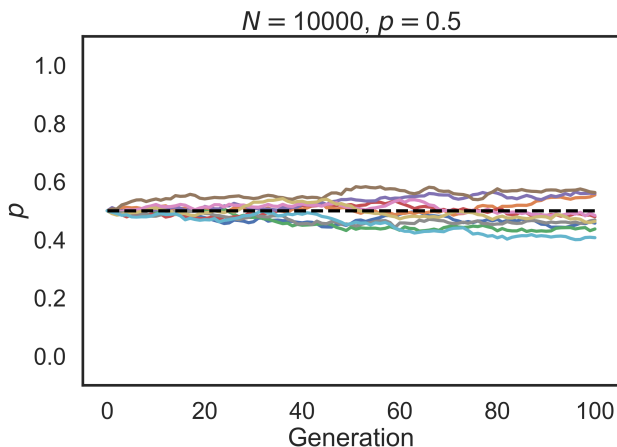
Evolution in finite populations

The shape of an evolutionary explanation



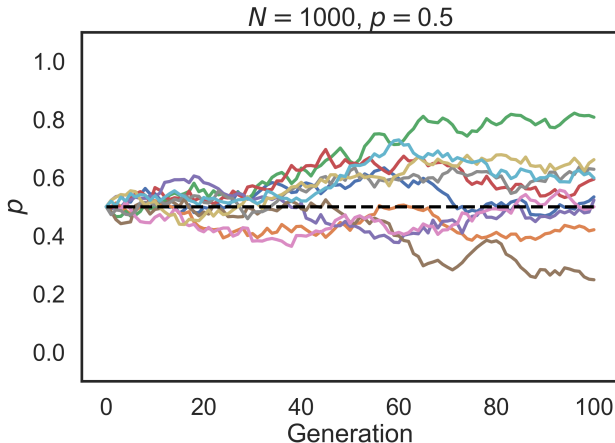
Evolution in finite populations

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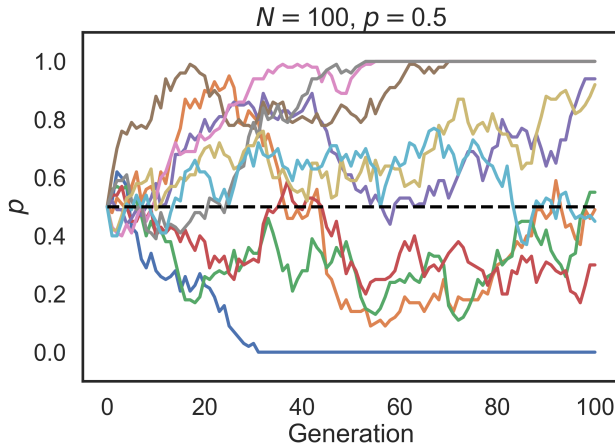
Evolution in finite populations

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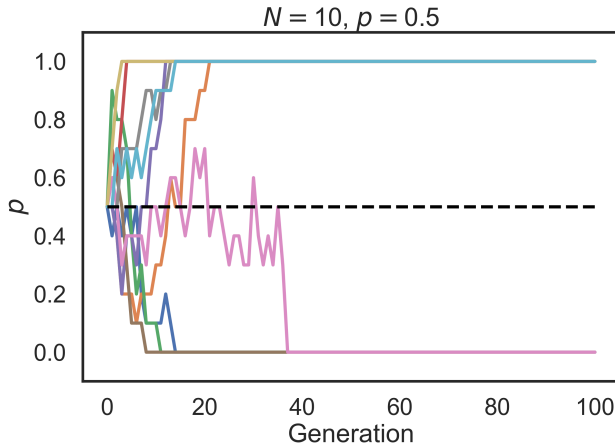
Evolution in finite populations

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Evolution in finite populations

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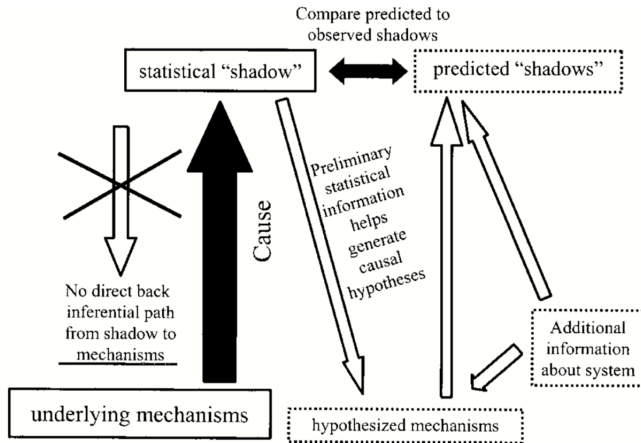
In search of an evolutionary cause



Pigliucci & Kaplan (2006) *Making sense of evolution*

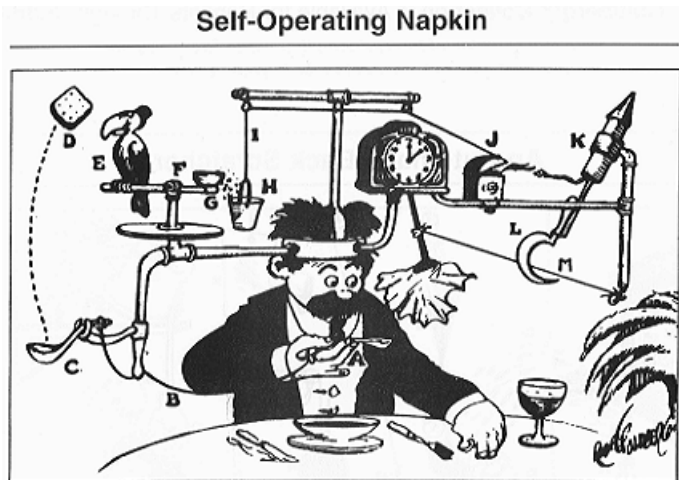
The shape of an evolutionary explanation

In search of an evolutionary cause



The shape of an evolutionary explanation

In search of an evolutionary cause



Rube Goldberg (1931) *Professor Butts and the Self-Operating Napkin*

The shape of an evolutionary explanation

In search of an evolutionary cause



Gould & Lewontin (1979) *The Spandrels of San Marco and the Panglossian Paradigm: A Critique of the Adaptationist Programme*

Outline

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The fingerprint of evolution

The utility of simulacra

The mind of *Caenorhabditis elegans*

The fingerprint of evolution

- ▶ A member of the phylum Nematoda.
- ▶ It grows to about 1 mm in length, and lives in the soil.
- ▶ 959 somatic cells in the hermaphrodite (302 neurons)

The mind of *Caenorhabditis elegans*

The fingerprint of evolution



Caenorhabditis elegans (Source: Wikipedia)

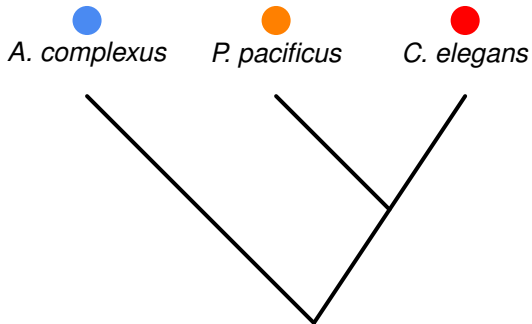
Comparative method

The fingerprint of evolution

- ▶ *P. pacificus* is usually found in association with different species of beetles.
- ▶ It grows to about 1 mm in length, and lives in the soil.
- ▶ 959 somatic cells in the hermaphrodite (302 neurons)

Comparative method

The fingerprint of evolution



Hong et al., *Evolution of neuronal anatomy and circuitry in two highly divergent nematode species*, eLife. 2019; 8: e47155.

Comparative method

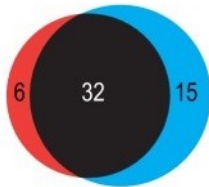
The fingerprint of evolution

- ▶ *P. pacificus* is usually found in association with different species of beetles.
- ▶ *C. elegans* is free-living and usually found on rotting fruit.
- ▶ The species shared their last common ancestor ≈ 100 mya.
- ▶ *Pristionchus* species show predatory behavior towards *C. elegans*.

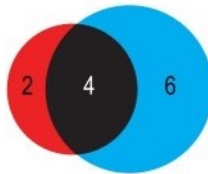
Comparative method

The fingerprint of evolution

Chemical connections



Gap junction connections

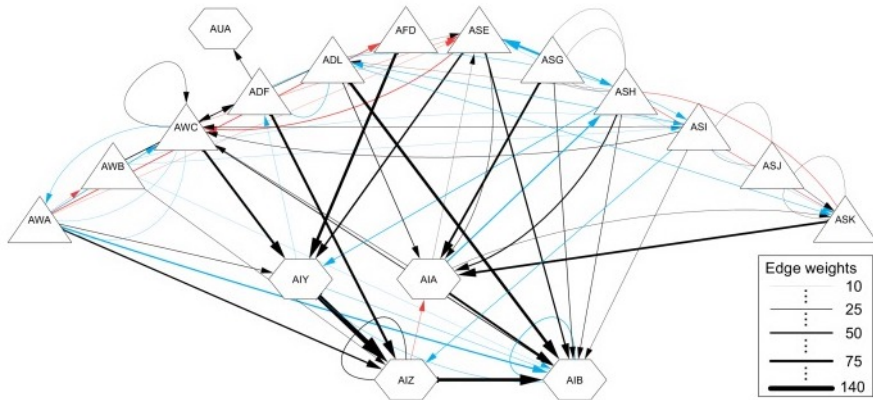


Conserved
P. pacificus
specific
C. elegans
specific

Hong et al., *Evolution of neuronal anatomy and circuitry in two highly divergent nematode species*, eLife. 2019; 8: e47155.

Comparative method

The fingerprint of evolution



Hong et al., *Evolution of neuronal anatomy and circuitry in two highly divergent nematode species*, eLife. 2019; 8: e47155.

Unweaving an evolutionary story

The fingerprint of evolution

Published online 11 November 2009 | Nature | doi:10.1038/news.2009.1079

News

Evolution of a single gene linked to language

Mutations in the *FOXP2* gene could help explain why humans can speak but chimps can't.

Kerri Smith

Two tiny changes in the sequence of one gene could have helped install the mechanisms of speech and language in humans.

In 2001, a gene called *FOXP2* was found to underlie a rare inherited speech and language disorder¹. It encodes a transcription factor called *FOXP2*, a protein 'dimmer-switch' that binds to DNA and helps to determine to what extent other genes are expressed as proteins.

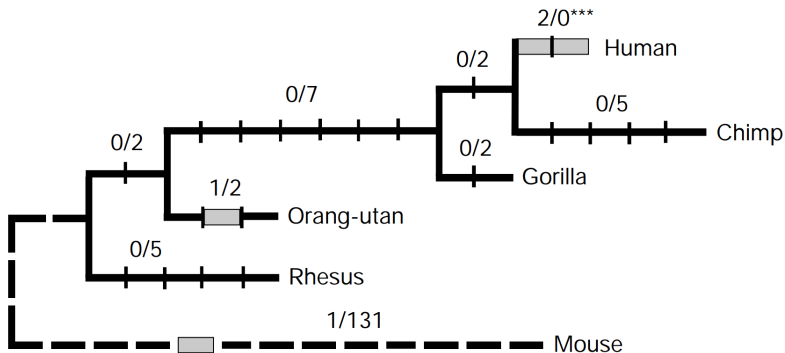


Changes to the sequence of one gene may help to explain why humans talk but chimps don't.

mlorenzphotography/ Getty

Unweaving an evolutionary story

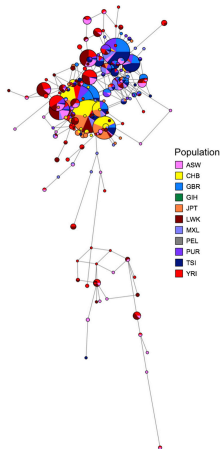
The fingerprint of evolution



Enard et al. (2002) *Molecular evolution of FOXP2, a gene involved in speech and language*. Nature Vol. 418

Unweaving an evolutionary story

The fingerprint of evolution



Atkinson et al., *No evidence for recent selection at FOXP2 among diverse human populations*, Cell. 2018 Sep 6; 174(6): 1424–1435.e15.

Outline

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The utility of simulacra

What is a simulacrum?

The utility of simulacra

simulacrum noun



Save Word

sim·u·la·crum | \,sim-yə-'la-krəm , -'lā- \

plural **simulacra** \,sim-yə-'la-kre , -'lā- \ *also* **simulacrums**

Definition of *simulacrum*

1 : IMAGE, REPRESENTATION

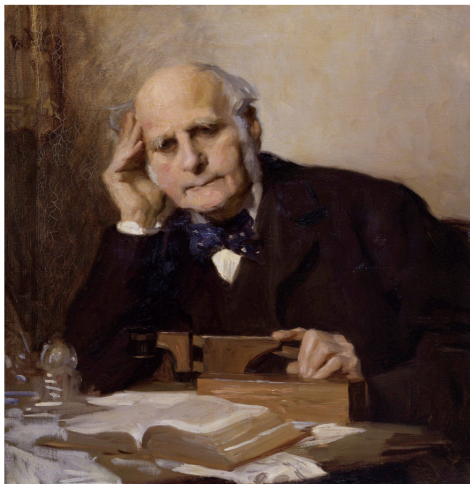
// a reasonable *simulacrum* of reality

— Martin Mayer

2 : an insubstantial form or semblance of something : TRACE

Understanding through a simulacrum

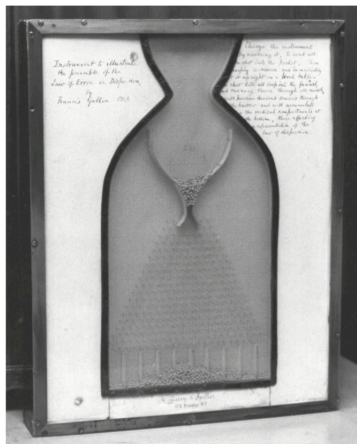
The utility of simulacra



Sir Francis Galton (1822 - 1911)

Typical laws of heredity (Feb 9, 1877)

The utility of simulacra

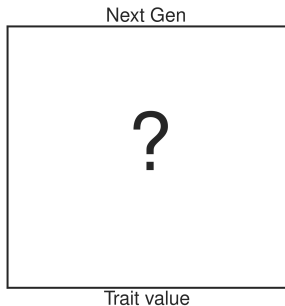
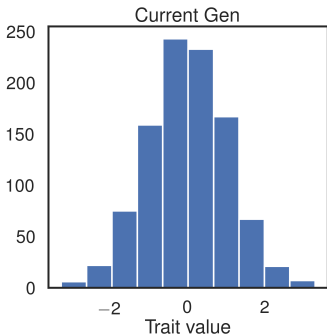


(a)

The bean machine, also known as the Galton Board or quincunx

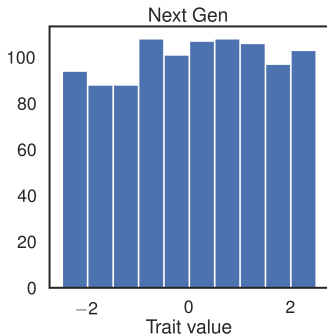
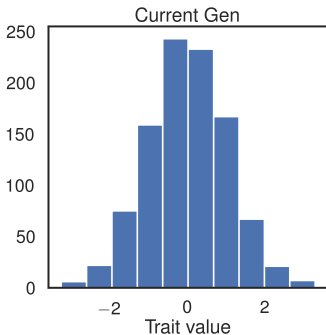
Typical laws of heredity (Feb 9, 1877)

The utility of simulacra



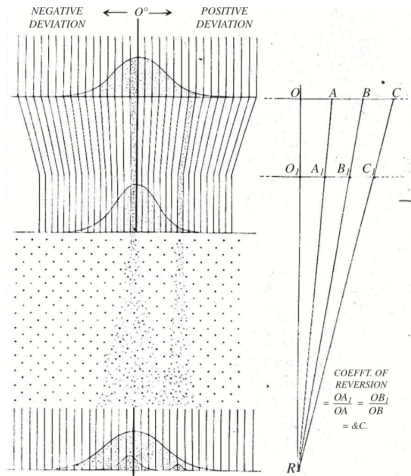
Typical laws of heredity (Feb 9, 1877)

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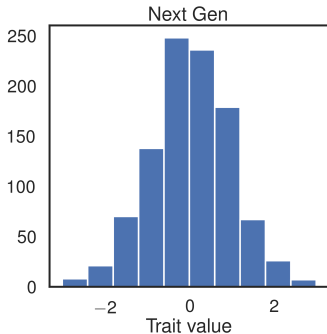
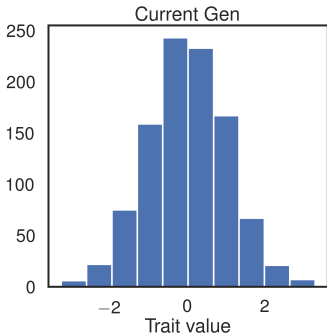
From a machine to a theory

The utility of simulacra



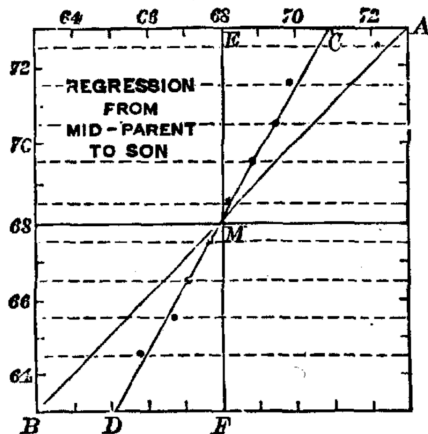
Typical laws of heredity (Feb 9, 1877)

The utility of simulacra



From a machine to a theory

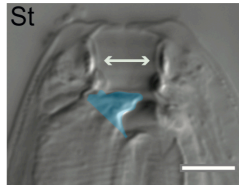
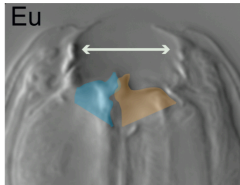
The utility of simulacra



Galton, *Natural Inheritance* (1889)

Modeling phenotypic plasticity

The utility of simulacra



Modeling phenotypic plasticity

The utility of simulacra

$$x_{St,J}^{t+1} = x_{St,J}^t + f\delta_x [x_{St,A}^t + x_{Eu,A}^t] - \lambda_x x_{St,J}^t - cx_{St,J}^t \sum_{i=x,y} i_{Eu,A}^t b_{x,i} (1 - \chi_i r_{x,i})$$

$$x_{St,A}^{t+1} = x_{St,A}^t + \lambda_x x_{St,J}^t - x_{St,A}^t \frac{f}{K} \sum_{i=x,y} i_A^t$$

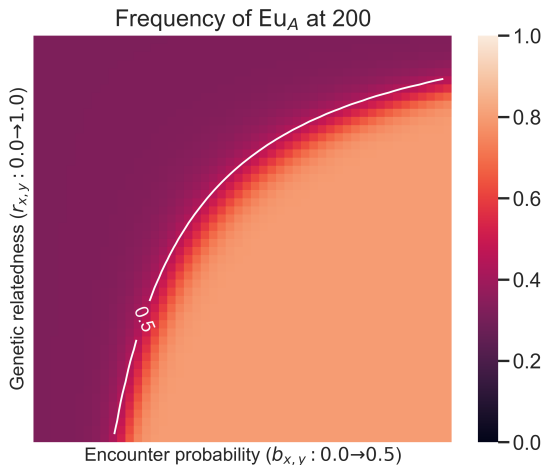
$$x_{Eu,J}^{t+1} = x_{Eu,J}^t + f(1 - \delta_x) [x_{Eu,A}^t + x_{St,A}^t] - \lambda_x x_{Eu,J}^t - cx_{Eu,J}^t \sum_{i=x,y} i_{Eu,A}^t b_{x,i} (1 - \chi_i r_{x,i})$$

$$x_{Eu,A}^{t+1} = x_{Eu,A}^t + \lambda_x x_{Eu,J}^t - x_{Eu,A}^t \frac{f}{K} \sum_{i=x,y} i_A^t \quad ,$$

Lightfoot et al., *Phenotypic plasticity shape Cannibalistic behaviours* (in prep)

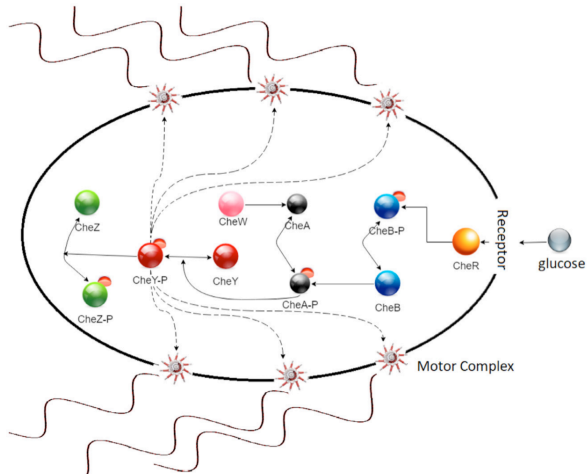
Modeling phenotypic plasticity

The utility of simulacra



Not too simple or too complex

The utility of simulacra



Vafadar et al., *From indication to decision: A hierarchical approach to model the chemotactic behavior of Escherichia coli*. Journal of Theoretical Biology 495 (2020) 110253.

Fin.



Neugebauer (1952) *The Exact Sciences in Antiquity*